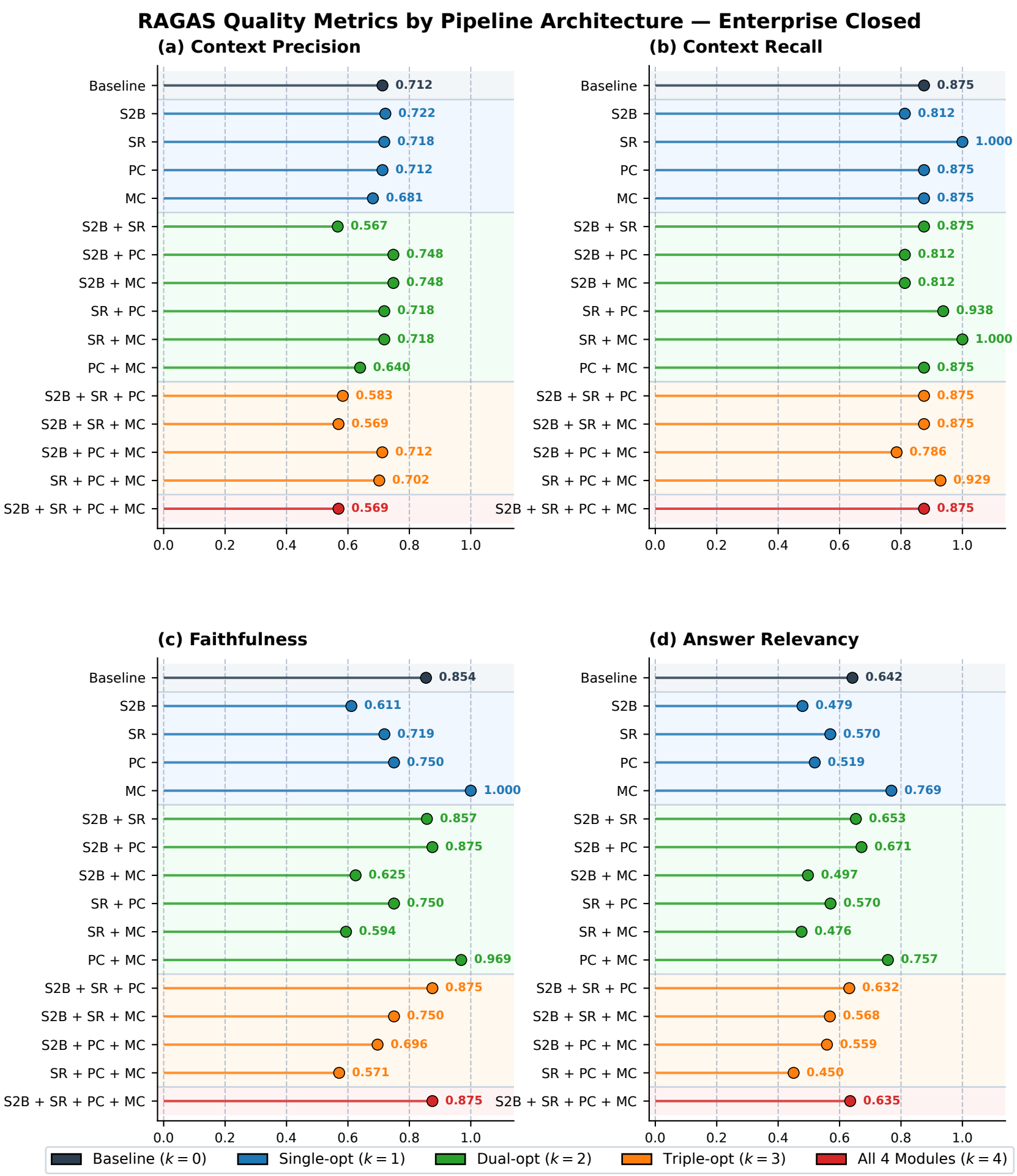
RAGAS Quality Metrics by Pipeline Architecture — Enterprise Closed
(a) Context Precision
Baseline 0.712
S2B 0.722
SR 0.718
PC 0.712
MC 0.681
S2B + SR 0.567
S2B + PC 0.748
S2B + MC 0.748
SR + PC 0.718
SR + MC 0.718
PC + MC 0.640
S2B + SR + PC 0.583
S2B + SR + MC 0.569
S2B + PC + MC 0.712
SR + PC + MC 0.702
S2B + SR + PC + MC 0.569
0.0 0.2 0.4 0.6 0.8 1.0
(b) Context Recall
Baseline 0.875
S2B 0.812
SR 1.000
PC 0.875
MC 0.875
S2B + SR 0.875
S2B + PC 0.812
S2B + MC 0.812
SR + PC 0.938
SR + MC 1.000
PC + MC 0.875
S2B + SR + PC 0.875
S2B + SR + MC 0.875
S2B + PC + MC 0.786
SR + PC + MC 0.929
S2B + SR + PC + MC 0.875
0.0 0.2 0.4 0.6 0.8 1.0
(c) Faithfulness
Baseline 0.854
S2B 0.611
SR 0.719
PC 0.750
MC 1.000
S2B + SR 0.857
S2B + PC 0.875
S2B + MC 0.625
SR + PC 0.750
SR + MC 0.594
PC + MC 0.969
S2B + SR + PC 0.875
S2B + SR + MC 0.750
S2B + PC + MC 0.696
SR + PC + MC 0.571
S2B + SR + PC + MC 0.875
0.0 0.2 0.4 0.6 0.8 1.0
(d) Answer Relevancy
Baseline 0.642
S2B 0.479
SR 0.570
PC 0.519
MC 0.769
S2B + SR 0.653
S2B + PC 0.671
S2B + MC 0.497
SR + PC 0.570
SR + MC 0.476
PC + MC 0.757
S2B + SR + PC 0.632
S2B + SR + MC 0.568
S2B + PC + MC 0.559
SR + PC + MC 0.450
S2B + SR + PC + MC 0.635
0.0 0.2 0.4 0.6 0.8 1.0
Baseline ($k = 0$)
Single-opt ($k = 1$)
Dual-opt ($k = 2$)
Triple-opt ($k = 3$)
All 4 Modules ($k = 4$)